External Database Integration Design — v3.1

> **This is one of the project's core commercial moats:** 90% of 50-100 person factories already use Dingxin / Chenghang / SuperBase / Excel. **"Can you read my legacy data?" is the #1 ERP procurement killer question.**
>
> Without this capability, every customer demo dies. With it, **a single customer is worth $30k/year**.

For full Chinese rationale and details see EXTERNAL_DB_INTEGRATION_DESIGN_ZH.md.

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1. Strategic Position

1.1 Why this matters more than mobile

Customer says	Real meaning	Without this we
"We use Dingxin now"	Won't replace it, you must read it	demo dies
"I have 500k legacy records"	Won't re-key, need migration	can't sign
"Connect to our SQL Server"	ODBC required	integrators block
"Customer PO comes in Excel"	Must ingest CSV / Excel	sales has no story

Migration anxiety is the top procurement killer. With external DB integration: "Your Dingxin doesn't need to die. We read it; AI auto-maps fields; you click ConfirmCard; done."

1.2 Same DNA as Conversational ERP

This is not a side track — it directly extends the "natural-language replaces training" promise:

> "Director Wang types: 'How much in Dingxin orders for May?' → AI cross-DB query" "Procurement types: 'Migrate Dingxin customers' → AI emits schema-mapping ConfirmCard"

Tool registry / RiskTier / ConfirmCard from Phase 1 are **all reused**.

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2. Legacy Systems Coverage

Customer's system	Connection	Priority	Difficulty
Dingxin Workflow ERP	SQL Server (pyodbc)	🔥 P0	⭐⭐
Chenghang ERP	SQL Server	🔥 P0	⭐⭐
SuperBase ERP	REST API (OAuth2)	🟠 P1	⭐⭐
SAP Business One	DI API / Service Layer REST	🟠 P1	⭐⭐⭐
Odoo	XML-RPC / REST	🟠 P1	⭐⭐
In-house Access / Excel	ODBC / openpyxl	🟢 PoC	⭐
CSV batches	Folder watcher	🟢 PoC	⭐
e-invoice SaaS	REST API	🟠 P1	⭐⭐
PLC / SCADA / MES	OPC UA / Modbus	⏸ P5+	⭐⭐⭐⭐⭐

MVP target: PoC = sqlite + csv. Phase 1.5 = + SqlServer + REST API.

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3. Four Connection Modes

```
┌──────────────────────────────────────────────────┐
│        LLM-ERP (Desktop Chat entry)                │
└──────────────────────────────────────────────────┘
       │
       │  AI tools:
       │   • query_external_db        (Federated)
       │   • migrate_from_external    (One-time)
       │   • sync_with_external       (Two-way)    Phase 2+
       │   • watch_external_changes   (CDC)         Phase 3+
       │
┌──────────────────────────────────────────────────┐
│  Connector Registry (@register_connector)         │
└──────────────────────────────────────────────────┘
       │
       │
       ▼
┌─────────────────┬──────────────────────────────┐
│ ① Federated     │ AI queries external, no copy │
│ ② One-time mig  │ Bulk import to LLM-ERP       │
│ ③ Two-way sync  │ Bidirectional + conflict     │ Phase 2
│ ④ CDC stream    │ Real-time event capture      │ Phase 3
└─────────────────┴──────────────────────────────┘
```

v3.0 scope: ① + ②. ③ ④ deferred pending customer feedback.

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4. Architecture

4.1 Code structure

```
backend/app/integrations/connectors/
├── __init__.py
├── base.py                  # Connector ABC + ConnectorMeta
├── registry.py              # @register_connector / get_connector
├── exceptions.py            # ConnectorError / TableNotFound / ...
├── sqlite_connector.py      # ✅ built-in (PoC)
├── csv_connector.py         # ✅ built-in (PoC)
├── postgres_connector.py    # Phase 1.5
├── sqlserver_connector.py   # Phase 1.5 (Dingxin / Chenghang)
├── mysql_connector.py       # Phase 1.5
├── rest_api_connector.py    # Phase 1.5
└── excel_connector.py       # Phase 1.5
```

4.2 Connector interface contract

```python
class Connector(ABC):
    meta: ConnectorMeta

    def __init__(self, config: dict): ...

    @abstractmethod
    async def test_connection(self) -> bool: ...

    @abstractmethod
    async def list_tables(self) -> list[str]: ...

    @abstractmethod
    async def query(
        self, table: str,
        filters: dict | None = None,
        limit: int = 100,
    ) -> list[dict]: ...

    async def schema_of(self, table: str) -> list[dict]: ...
```

4.3 Four security defenses

Layer	Defense	Why
1	**read-only by default**	Never accidentally write to legacy system
2	**table whitelist**	`query(table)` must pass `list_tables()` check
3	**parameterized filters**	No SQL string concatenation, prepared statements only
4	**audit log**	Every external query logged with connection / table / filter

5. AI Tool Specs

5.1 PoC phase (v3.0) — 3 read tools

Tool	RiskTier	Description
`list_external_connections`	READ	List configured external DB connections
`list_external_tables`	READ	List tables in a connection
`query_external_db`	READ	Cross-DB query with filters + limit

5.2 Phase 1.5 — add 4 more

- `register_external_connection_with_confirm` (HARD_WRITE)
- `delete_external_connection_with_confirm` (HARD_WRITE)
- `preview_schema_mapping` (READ — AI auto-suggests field mappings)
- `migrate_from_external_with_confirm` (HARD_WRITE — one-time bulk import)

6. Killer Conversational Scenarios

Scenario A: Federated query

```
Wang: "How much in Dingxin orders for May?"
AI: list_external_connections → finds legacy_dingxin
    list_external_tables(legacy_dingxin) → finds OrderHeader
    query_external_db(legacy_dingxin, OrderHeader,
                      filters={order_date_gte: 2026-05-01})
    Sums Amount → "$3.2M (45 orders) in Dingxin; $580K (12 orders) in LLM-ERP."
```

Scenario B: One-time migration

```
Alice: "Migrate customers from Dingxin"
AI: list_external_tables → Customer
    preview_schema_mapping → auto suggests:
      Dingxin.Customer.CustNo   → LLM-ERP.Customer.code
      Dingxin.Customer.CustName → LLM-ERP.Customer.name
      Dingxin.Customer.Grade    → LLM-ERP.Customer.grade
    Emits ConfirmCard: "Import 124 customers. Conflict policy: overwrite by code."
    Alice clicks Confirm → migration runs → progress bar → done.
```

Scenario C: CSV folder watch

```
Sales: "Customer A drops PO CSVs in D:/orders daily at 9am"
AI: register_external_connection(name=customer_a_csv, connector=csv_folder,
                                  config={folder: D:/orders})
    schedule_external_sync(every=5min, mode=ingest_new_files,
                           target=sales_order)
    "Set up. Every 5 minutes scans D:/orders for new CSVs and imports as SO."
```

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7. Schema Mapping AI

7.1 Auto-suggestion logic

1. **Column name match** (exact + LLM fuzzy via Glossary)
2. **Type inference** (`VARCHAR(50)` ↔ String, `DECIMAL(18,2)` ↔ Float)
3. **Domain constraints** (LLM-ERP `customer.grade` ∈ {A,B,C,D} → warn)
4. **Confidence score**:
    - 95%+ → auto apply
    - 70-95% → emit ConfirmCard for human review
    - <70% → ask "no match found, create new field?"

7.2 Reuses Phase 2 Glossary

The Glossary from `CONVERSATIONAL_ERP_DESIGN_EN.md` directly applies:

- "Customer" / "CustNo" / "Cust" → all map to LLM-ERP.Customer
- "Part" / "Material" / "Item" → all map to LLM-ERP.Part

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8. Phase Plan

Phase 1.5 (parallel with Phase 1, 2 weeks)

#	Task	Days	GAP
1.5.1	Connector framework (base + registry)	0.5	G-501
1.5.2	SqliteConnector + CsvFolderConnector (PoC)	0.5	G-502
1.5.3	3 read tools (list / list_tables / query)	0.5	G-503
1.5.4	Smoke tests (PoC)	0.5	G-504
1.5.5	SqlServerConnector (pyodbc)	1	G-505
1.5.6	Postgres + MySQL connectors	1	G-506
1.5.7	RestApi connector + SuperBase / SAP B1 profiles	2	G-507
1.5.8	Schema mapping AI	2	G-508
1.5.9	migrate_from_external_with_confirm	1.5	G-509
1.5.10	Customer system profiles (Dingxin / Chenghang)	1	G-510

Total: ~10.5 work days, parallel with Phase 1.

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9. Risks & Mitigations

Risk	Impact	Mitigation
Driver install pain (SQL Server pyodbc)	IT frustration	Pre-installed in Docker; `install.sh` one-click
Legacy schemas vary wildly	Mapping AI errors	Pre-built profiles for common systems + human final confirm
Accidental writes to legacy	Customer data corruption	read-only default; hard-write requires ConfirmCard
SQL injection	Security incident	Table whitelist + parameterized statements only
Slow federated queries	Bad UX	Default limit 100; big queries → background job + Email

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10. Sales Story

One-liner

> "Your Dingxin doesn't have to die. We read it. AI migrates gradually. Zero risk during transition."

Three-liner

> "ERP buyers' #1 fear is legacy data. We connect directly to Dingxin / Chenghang / SuperBase / SAP B1 / Excel / CSV. AI auto-maps fields; you click a ConfirmCard; done — no SQL required.
>
> 90% of competitors want you to 'kill the old system first.' We say 'run both for 6 months if you want.' **Zero-risk transition**."

Last updated: 2026-05-15 (v3.1 strategic supplement: External DB integration) **Related GAPs**: G-501 ~ G-510 **Roadmap**: Phase 1.5